

With vibe coding implemented for Selenium/C#

Prompt.js:

```
SELENIUM_CSHARP_PAGE_ONLY: `
  Instructions:
  - Generate ONLY a Selenium C# Page Object Class (no test code).
  - Add XMLDoc for methods & class.
  - Use Selenium 2.30+ compatible imports.
  - Use meaningful method names.
  - Do NOT include explanations or test code.

  Context:
  DOM:
  \\\`html
  \${domContent}
  \\\`

  Example:
  \\\`csharp
  using OpenQA.Selenium;

  /**
   * Page Object for Component Page
   */
  public class ComponentPage {
    // Add methods as per the DOM
  }
  \\\`

  Persona:
  - Audience: Automation engineer focusing on maintainable POM structure.

  Output Format:
  - A single C# class inside a \\\`csharp\\\` block.

  Tone:
  - Clean, maintainable, enterprise-ready.
  `,

  /**
   * Cucumber Feature File Only Prompt
   */
  CUCUMBER_ONLY: `
  Instructions:
```

- Generate ONLY a Cucumber (.feature) file.
- Use Scenario Outline with Examples table.
- Make sure every step is relevant to the provided DOM.
- Do not combine multiple actions into one step.
- Use South India realistic dataset (names, addresses, pin codes, mobile numbers).
- Use dropdown values only from provided DOM.
- Generate multiple scenarios if applicable.

Context:

DOM:

```

\`\`\`html
\${domContent}
\`\`\`

```

Example:

```

\`\`\`gherkin

```

Feature: Login to OpenTaps

Scenario Outline: Successful login with valid credentials

```

Given I open the login page
When I type "<username>" into the Username field
And I type "<password>" into the Password field
And I click the Login button
Then I should be logged in successfully

```

Examples:

username	password
"testuser"	"testpass"
"admin"	"admin123"

```

\`\`\`

```

Persona:

- Audience: BDD testers who only need feature files.

Output Format:

- Only valid Gherkin in a `\`\`\`gherkin\`\`\`` block.

Tone:

- Clear, structured, executable.

` ,

/**

* Cucumber with Step Definitions (C#)

*/

CUCUMBER_WITH_SELENIUM_CSHARP_STEPS: `

Instructions:

- Generate BOTH:
 1. A Cucumber .feature file.
 2. A C# step definition class for Selenium.
- Do NOT include Page Object code.
- Step defs must include WebDriver setup, explicit waits, and actual Selenium code.
- Use SpecFlow and NUnit framework.
- Use Scenario Outline with Examples table (South India realistic data).

Context:

DOM:

\\`\\`\\`html

\\\${domContent}

\\`\\`\\`

URL: \\\${pageUrl}

Example:

\\`\\`\\`gherkin

Feature: Login to OpenTaps

Scenario Outline: Successful login with valid credentials

Given I open the login page
When I type "<username>" into the Username field
And I type "<password>" into the Password field
And I click the Login button
Then I should be logged in successfully

Examples:

username password
"admin" "admin123"
"testuser" "testpass"

\\`\\`\\`

\\`\\`\\`csharp

```
using OpenQA.Selenium;  
using OpenQA.Selenium.Chrome;  
using OpenQA.Selenium.Support.UI;  
using TechTalk.SpecFlow;  
using NUnit.Framework;  
using System;  
using SeleniumExtras.WaitHelpers;
```

namespace LeafTaps.StepDefinitions

```

{
    /// <summary>
    /// Step definitions for Login feature
    /// </summary>
    [Binding]
    public class LoginStepDefinitions
    {
        private IWebDriver driver;
        private WebDriverWait wait;

        /// <summary>
        /// Setup method executed before each scenario
        /// </summary>
        [Before]
        public void SetUp()
        {
            driver = new ChromeDriver();
            wait = new WebDriverWait(driver, TimeSpan.FromSeconds(10));
            driver.Manage().Window.Maximize();
        }

        /// <summary>
        /// Teardown method executed after each scenario
        /// </summary>
        [After]
        public void TearDown()
        {
            if (driver != null)
            {
                driver.Quit();
            }
        }

        /// <summary>
        /// Opens the login page
        /// </summary>
        [Given(@"I open the login page")]
        public void OpenLoginPage()
        {
            driver.Navigate().GoToUrl(@"\${pageUrl}");
        }

        /// <summary>
        /// Enters username into the Username field
        /// </summary>

```

```

        [When(@"I type \"(.*)\" into the Username field")]
        public void EnterUsername(string username)
        {
            IWebElement element =
wait.Until(ExpectedConditions.ElementToBeClickable(By.Id("username")));
            element.SendKeys(username);
        }

        /// <summary>
        /// Enters password into the Password field
        /// </summary>
        [When(@"I type \"(.*)\" into the Password field")]
        public void EnterPassword(string password)
        {
            IWebElement element =
wait.Until(ExpectedConditions.ElementToBeClickable(By.Id("password")));
            element.SendKeys(password);
        }

        /// <summary>
        /// Clicks the Login button
        /// </summary>
        [When(@"I click the Login button")]
        public void ClickLogin()
        {
            IWebElement loginButton =
driver.FindElement(By.XPath("//button[contains(text(),'Login')]"));
            loginButton.Click();
        }

        /// <summary>
        /// Verifies successful login
        /// </summary>
        [Then(@"I should be logged in successfully")]
        public void VerifyLogin()
        {
            IWebElement successElement =
wait.Until(ExpectedConditions.VisibilityOfAllElementsLocatedBy(By.ClassName("succ
ess")))[0];
            Assert.IsTrue(successElement.Displayed, "Login was not
successful");
        }
    }
}

```

Persona:

- Audience: QA engineers working with Cucumber & Selenium in C#.

Output Format:

- Gherkin in ````gherkin```` block + C# code in ````csharp```` block.

Tone:

- Professional, executable, structured, following C# conventions.

```
export const CODE_GENERATOR_TYPES = {
  SELENIUM_JAVA_PAGE_ONLY: 'Selenium-Java-Page-Only',
  SELENIUM_CSHARP_PAGE_ONLY: 'Selenium-CSharp-Page-Only',
  CUCUMBER_ONLY: 'Cucumber-Only',
  CUCUMBER_WITH_SELENIUM_JAVA_STEPS: 'Cucumber-With-Selenium-Java-Steps',
  CUCUMBER_WITH_SELENIUM_CSHARP_STEPS: 'Cucumber-With-Selenium-CSharp-Steps',
  PLAYWRIGHT_TYPESCRIPT_POM: 'Playwright-TypeScript-POM',
  PLAYWRIGHT_TYPESCRIPT_SPEC: 'Playwright-TypeScript-Spec',
  PLAYWRIGHT_TYPESCRIPT_POM_AND_SPEC: 'Playwright-TypeScript-POM-And-Spec',
};
```

Chat.js:

```
if (!isFeatureChecked && !isPageChecked) {
  console.warn('No generation mode selected. Defaulting to Page Object generation.');
```

// Default fallback to page object generation

```
  if (this.isJavaSelenium(lang, eng)) {
    promptKeys.push('SELENIUM_JAVA_PAGE_ONLY');
  } else if (this.isTypeScriptPlaywright(lang, eng)) {
    promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM');
  } else if (this.isCSharpSelenium(lang, eng)) {
    promptKeys.push('SELENIUM_CSHARP_PAGE_ONLY');
  }
  return promptKeys;
}
```

```
if (isFeatureChecked && isPageChecked) {
  // Both feature and page selected - generate combined output
  if (this.isJavaSelenium(lang, eng)) {
    promptKeys.push('CUCUMBER_WITH_SELENIUM_JAVA_STEPS');
```

} else if (this.isTypeScriptPlaywright(lang, eng)) {

```

        promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM_AND_SPEC');
    } else if (this.isCSharpSelenium(lang, eng)) {
        promptKeys.push('CUCUMBER_WITH_SELENIUM_CSHARP_STEPS');
    }
    else {
        // For non-Java/Selenium and non-TypeScript/Playwright
combinations
        promptKeys.push('CUCUMBER_ONLY');
        this.addUnsupportedLanguageMessage(lang, eng);
    }
}

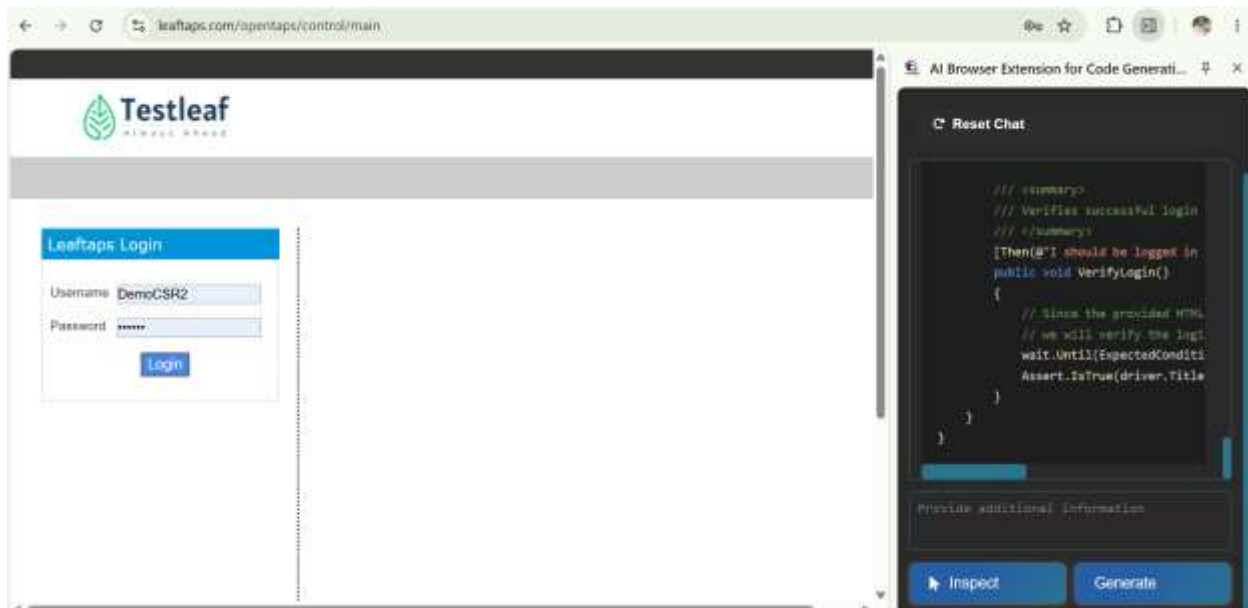
```

```

if (isFeatureChecked && isPageChecked) {
    // Both feature and page selected - generate combined output
    if (this.isJavaSelenium(lang, eng)) {
        promptKeys.push('CUCUMBER_WITH_SELENIUM_JAVA_STEPS');
    } else if (this.isTypeScriptPlaywright(lang, eng)) {
        promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM_AND_SPEC');
    } else if (this.isCSharpSelenium(lang, eng)) {
        promptKeys.push('CUCUMBER_WITH_SELENIUM_CSHARP_STEPS');
    }
    else {
        // For non-Java/Selenium and non-TypeScript/Playwright
combinations
        promptKeys.push('CUCUMBER_ONLY');
        this.addUnsupportedLanguageMessage(lang, eng);
    }
} else if (isFeatureChecked) {
    // Feature file only
    if (this.isTypeScriptPlaywright(lang, eng)) {
        promptKeys.push('PLAYWRIGHT_TYPESCRIPT_SPEC');
    } else {
        promptKeys.push('CUCUMBER_ONLY');
    }
} else if (isPageChecked) {
    // Page object only
    if (this.isJavaSelenium(lang, eng)) {
        promptKeys.push('SELENIUM_JAVA_PAGE_ONLY');
    } else if (this.isTypeScriptPlaywright(lang, eng)) {
        promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM');
    } else if (this.isCSharpSelenium(lang, eng)) {
        promptKeys.push('SELENIUM_CSHARP_PAGE_ONLY');
    }
    else {
        this.addUnsupportedLanguageMessage(lang, eng);
    }
}

```

```
    }
}
```



Feature: Login to OpenTaps with South India realistic data

Scenario Outline: Successful login with valid credentials

Given I open the login page
 When I type "<username>" into the Username field
 And I type "<password>" into the Password field
 And I click the Login button
 Then I should be logged in successfully

Examples:

username	password
"ramesh123"	"chennai123"
"suresh456"	"bangalore456"
"kumar789"	"hyderabad789"

```
using OpenQA.Selenium;
using OpenQA.Selenium.Chrome;
using OpenQA.Selenium.Support.UI;
using TechTalk.SpecFlow;
using NUnit.Framework;
using System;
using SeleniumExtras.WaitHelpers;

namespace LeafTaps.StepDefinitions
{
    /// <summary>
    /// Step definitions for Login feature
    /// </summary>
    [Binding]
    public class LoginStepDefinitions
    {

```



```

private IWebDriver driver;
private WebDriverWait wait;

/// <summary>
/// Setup method executed before each scenario
/// </summary>
[Before]
public void SetUp()
{
    driver = new ChromeDriver();
    wait = new WebDriverWait(driver, TimeSpan.FromSeconds(10));
    driver.Manage().Window.Maximize();
}

/// <summary>
/// Teardown method executed after each scenario
/// </summary>
[After]
public void TearDown()
{
    if (driver != null)
    {
        driver.Quit();
    }
}

/// <summary>
/// Opens the login page
/// </summary>
[Given(@"I open the login page")]
public void OpenLoginPage()
{
    driver.Navigate().GoToUrl("https://leaftaps.com/opentaps/control/main");

    /// <summary>
    /// Enters username into the Username field
    /// </summary>
    [When(@"I type "(.*)"" into the Username field")]
    public void EnterUsername(string username)
    {
        IWebElement element =
        wait.Until(ExpectedConditions.ElementToBeClickable(By.Id("username")));
        element.SendKeys(username);
    }

    /// <summary>
    /// Enters password into the Password field
    /// </summary>
    [When(@"I type "(.*)"" into the Password field")]
    public void EnterPassword(string password)
    {
        IWebElement element =
        wait.Until(ExpectedConditions.ElementToBeClickable(By.Id("password")));
        element.SendKeys(password);
    }
}

```

```

    /// <summary>
    /// Clicks the Login button
    /// </summary>
    [When(@"I click the Login button")]
    public void ClickLogin()
    {
        IWebElement loginButton =
driver.FindElement(By.XPath("//input[@value='Login']"));
        loginButton.Click();
    }

    /// <summary>
    /// Verifies successful login
    /// </summary>
    [Then(@"I should be logged in successfully")]
    public void VerifyLogin()
    {
        // Since the provided HTML does not contain a success message,
        // we will verify the login by checking the title of the page
        wait.Until(ExpectedConditions.TitleIs("Login Success"));
        Assert.IsTrue(driver.Title.Contains("Login Success"), "Login was
not successful");
    }
}

```